GRIDLOCK – ADVANCE MULTI-LAYERED GRID BASED AUTHENTICATION SYSTEM

GridLock is a next-generation, multi-layer authentication system that secures logins through customizable grid patterns combining text inputs and color selections. Users can adjust grid size for desired security levels, while encrypted and hashed storage ensures complete data protection. Seamlessly integrable via OIDC/OAuth, GridLock offers a flexible, secure, and user-friendly alternative to traditional passwords.

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Grid Lock

INTRODUCTION

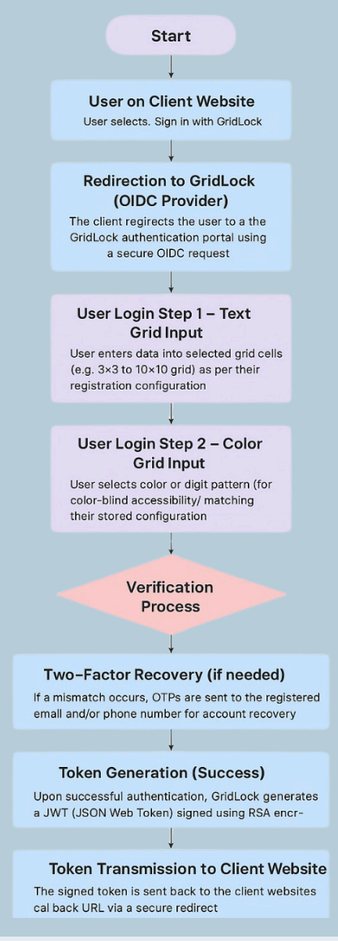
Most websites still use simple text passwords that are easy to guess, reuse, or steal can be bypass by phishing and brute force techniques etc. As digital use grows, these weak passwords make accounts and data vulnerable to attacks like brute force and phishing.
Problems with Existing Solutions

- Weak & Reused Passwords
- Hard to balance security & use
- Vulnerable to phishing/shoulder-surfing

OBJECTIVE

- Innovative
- Multilevel Authentication System
- Highly Secure Authentication with just a few clicks
- User-Friendly
- Flexible Difficulty
- The Entropy of System depends upon the User

METHODOLOGY



RESULTS

GridLock successfully implements a revolutionary authentication framework that transcends traditional username-password systems by introducing grid-based, multi-factor verification. Through its two-step process of text and color/digit grids, it delivers unprecedented entropy and resistance against brute-force, phishing, and data breach attacks.
By leveraging OIDC integration and secure token exchange, GridLock provides seamless compatibility with modern web platforms while maintaining user privacy and data integrity.
With its visually intuitive interface and cryptographically protected credentials, GridLock:

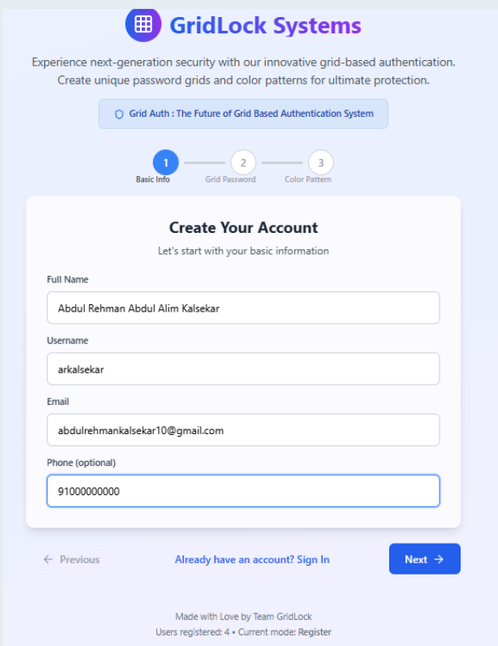
- Offers hyper-secure, personalized authentication for every user
- Eliminates password fatigue and reduces reset dependencies
- Strengthens protection through layered encryption and salting
- Enables fast, privacy-centric logins across multiple websites

In essence, GridLock transforms the way users interact with digital security — combining usability, adaptability, and state-of-the-art protection to redefine the future of authentication.

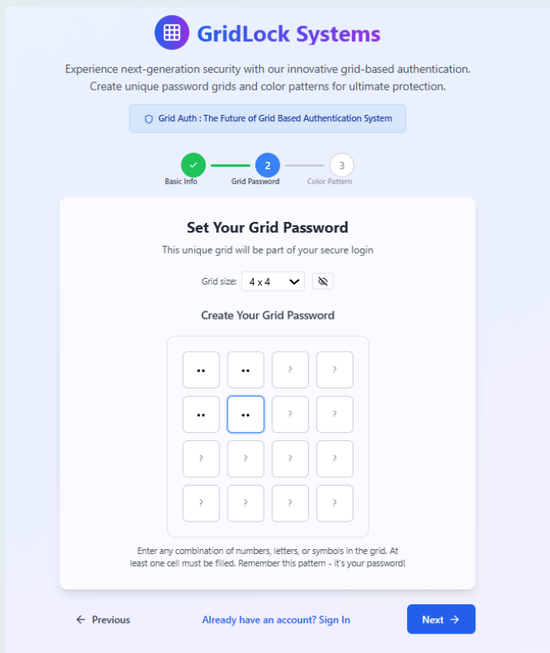
ANALYSIS

The analysis of the GridLock authentication system focuses on evaluating its security performance, usability, and efficiency compared to traditional password-based methods. Through entropy calculation, brute-force resistance testing, and user experience simulation, GridLock demonstrates a significant advancement in secure digital authentication.

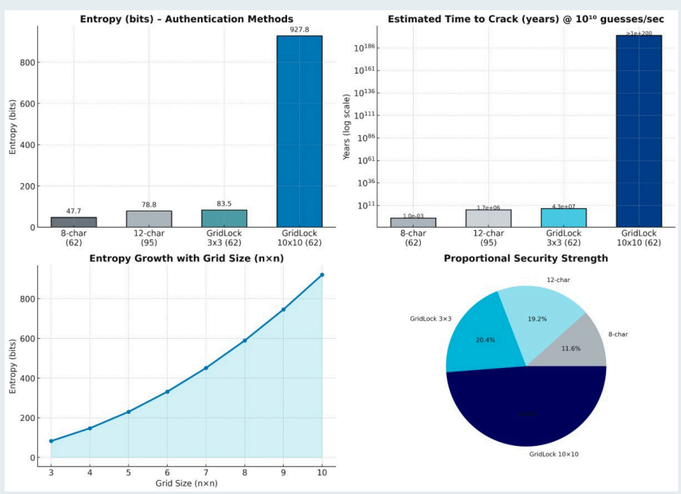
- High Entropy and Resistance to Attacks.
- Usability and Recall Efficiency.
- System Efficiency and Integration.



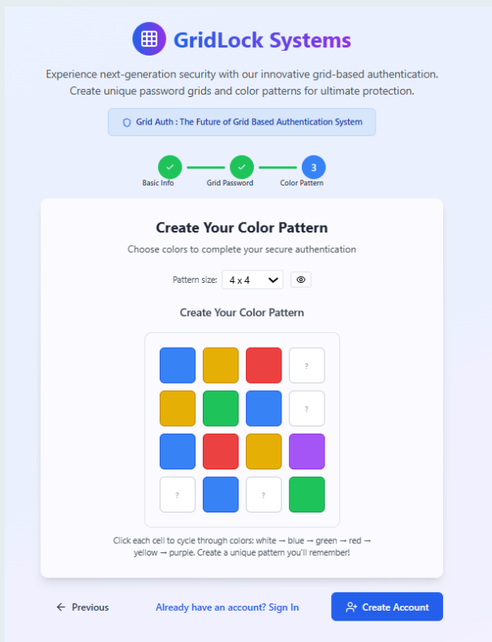
USER DATA COLECTION



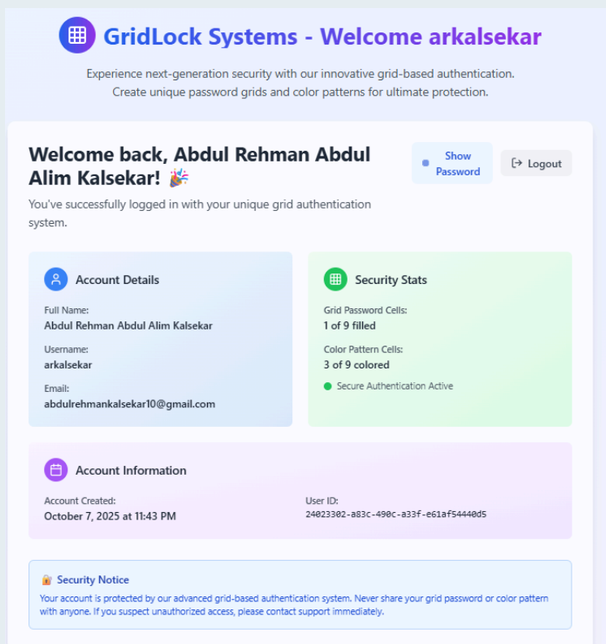
LAYER – 1



GRAPHICAL ANALYSIS



LAYER – 2



DASHBOARD